

Invoice Pipeline — Prototype Plan v3

Aligned with Design Reference v7 — Dynamics 365 CE, binary transport, Function decision engine, no duplicate-invoice verification.

Prepared: 18 June 2026 · Duration: ~10 working days · Supersedes v2

Changes from prototype plan v2

v2 was aligned to Design Reference v5 (three business categories, Blob+SAS transport, a Dataverse semantic dedup query). v3 realigns to Design Reference v7 and folds in every architecture decision since: binary transport, an Azure Function decision engine, item-id idempotency, and — most recently — the removal of duplicate-invoice verification.

#	Change	Driven by
1	CU transport is now BINARY (begin_analyze_binary) from the Azure Function. No Blob staging container and no SAS lifecycle.	CU cannot authenticate to SharePoint; binary avoids anonymous links
2	New component: an Azure Function decision engine. Gates A1/B2/B3/B4/GST and the CU call live in Python, not Logic App expressions.	Testable routing logic; one place for the gates
3	Document identity is the SharePoint item UniqueId, not a SHA-256 byte hash. InvoiceHashStore becomes InvoiceExtractProcessLog (a processing ledger). A1 is an atomic idempotency claim.	Stable per-ingestion identity; idempotency over hashing
4	Duplicate-invoice verification REMOVED: the Phase 6 Dataverse dup query (B5) and the alternate key are dropped. The happy path writes directly.	Requirement change — no duplicate-invoice check
5	Router has two categories (general_invoice, other). Property tax is deferred — no custom tax schema is built in the prototype.	Property tax analyzer deferred to post-prototype
6	General invoice analyzer is built on prebuilt-document, not prebuilt-invoice.	prebuilt-invoice is not a valid base analyzer
7	Routing gate B1 dropped (no per-category classifier confidence at GA); a GST reconciliation gate is added (total minus GST must reconcile to 5%).	GA response has no classifier confidence; GST control
8	Sample set drops the manual-entry-duplicate case and dedicated property tax bills. The processing ledger replaces the byte-hash store in Phase 3.	No dedup test; property tax deferred

Progress as of 18 June

Several phases are already substantially complete; v3 is a live plan, not a fresh start.

Phase	Status	Evidence
1 Azure foundation	Substantially complete	Foundry resource, storage, Key Vault, managed identity and RBAC provisioned in West US; gpt-5.2 deployed
2 CU validation	Substantially complete	generalinvoice and invoicrouter analyzers created and exercised; accuracy scorecard harness in place; B1 concluded absent (dropped), GST gate validated
3 Processing ledger	Complete	InvoiceExtractProcessLog validated by the ledger harness: item-id keying, idempotent upsert, atomic claim
4 Function + Logic App	Function code written and offline-validated; Logic App not built	Decision-engine project (A1 atomic claim, binary CU call, gates) passes offline routing and A1 tests
5–7	Not started	Review queue, Dynamics write-back, end-to-end run remain

Timeline overview

Phase	Focus	Days	Key output	Gate
Pre-work	Sample docs + CE prep	Before D1	12 labelled docs; CE admin answers	—
Phase 1	Azure foundation	Day 1	All resources in West US; ledger table	Gate 1
Phase 2	CU validation	Days 2–4	Accuracy scorecard; router + general analyzer	Gate 2
Phase 3	Processing ledger	Day 4 pm	Item-id idempotency working	—
Phase 4	Function + Logic App skeleton	Days 5–7	Function deployed; routing branches working	—
Phase 5	Review queue	Day 8	Queue + approval flow working	—
Phase 6	Dynamics CE write-back	Days 8–9	Dataverse rows created (no dup query)	—
Phase 7	End-to-end accuracy run	Days 9–10	Findings doc + go/no-go	Gate 3 / Final

Pre-work — documents and CE preparation

A. Sample document collection (12 documents)

Document type	Count	Selection criteria
General commercial invoices	5	Five most structurally different vendors: utility, tradesperson, insurer, landscaper, one-page emailed invoice
Handwritten general invoices	2	Real handwritten receipts — must drive is_handwritten = yes and route to review (B3)
Scanned general invoices	2	Photographed or scanner-produced — must classify but is_handwritten = no
Multi-page invoice	1	Line items spanning pages — tests the known CU table bug and the GST reconciliation gate
Non-invoice documents	2	Contract + statement — validate the 'other' category (B2 reject)

Record per document: vendor_name, invoice_number, invoice_date, gst_amount, total_invoice_amount, and service_address (the six critical fields), plus payment_due_date and line-item count where applicable. Include at least one invoice with a deliberate GST/total inconsistency to exercise the GST gate.

Dropped from v2: the manual-entry-duplicate document (no duplicate-invoice verification) and the dedicated property tax bills (property tax analyzer deferred). Property tax samples are collected when that analyzer is built post-prototype.

B. CE preparation — answers needed from the CE administrator

Item	Why needed	Blocks
Sandbox environment URL	All prototype writes go to sandbox, never production	Phase 6
Invoice table logical name	Dataverse 'Add a new row' targets it	Phase 6
Field logical names (invoicenum, totalamount, duedate, name/vendor)	Canonical field mapping	Phase 6
Vendor/account lookup model	Does the invoice row require an account lookup? Is 'find or create account' needed before the write?	Phase 6 — largest unknown
Service account or managed identity for Dataverse	Logic App authentication to Dataverse (managed identity preferred for production)	Phase 6

Schedule risk. Request the CE answers on Day 1 — they are not needed until Phase 6, but the account-lookup question may require a data-model discussion that takes days to resolve. The v2 alternate-key question is withdrawn: there is no duplicate-enforcement key. The v2 'instalment 2 handling' question is withdrawn with property tax.

Phase 1 — Azure foundation (Day 1)

Status. Substantially complete; steps below are the checklist to confirm against the existing dev resource group.

Step 1.1 — Resource group and region

- West US, single resource group. Canada Central is incompatible — do not deviate.

Step 1.2 — Foundry resource

- Standard S0; note the endpoint URL. The Content Understanding endpoint and key live here.

Step 1.3 — Storage account

- Standard / LRS, West US. Create the Table InvoiceExtractProcessLog. **No Blob container** — the binary transport sends bytes straight to CU; there is no landing zone to stage.
- The Function also needs an AzureWebJobsStorage account; this same account can serve.

Step 1.4 — Deploy the LLM model

- Deploy gpt-5.2 (not gpt-4.1, which retires Oct 2026).

Step 1.5 — Set the CU default model deployment

- PATCH the contentunderstanding defaults so the analyzers resolve gpt-5.2; expect HTTP 200.

Step 1.6 — Identity and RBAC

- User-assigned managed identity with Cognitive Services User on the Foundry resource and Storage Table Data Contributor on the storage account. Set AZURE_CLIENT_ID to disambiguate the identity.

Success criteria

- All resources in the same resource group and region (West US).
- Defaults PATCH returns HTTP 200; Content Understanding Studio shows the resource.
- InvoiceExtractProcessLog table exists; managed-identity roles assigned. No Blob container required.
- CE prep questions sent to the CE administrator.

Phase 2 — Content Understanding validation (Days 2–4)

Status. Substantially complete via the CU test harness; the steps below are the validation record.

Step 2.1 — Studio smoke test

- Run the 5 clean digital invoices through the general invoice analyzer; record fields, values, confidence; compare to the manual baseline.

Step 2.2 — Create the analyzers

Analyzer ID	baseAnalyzerId	enableOc r	Schema notes
generalinvoice	prebuilt-document	true	Invoice fields plus is_handwritten, vendor_category, anomaly_flag (classify/generate)
invoicerouter	prebuilt-document	true	Two contentCategories: general_invoice (→ generalinvoice) and other

- Constraints learned: analyzer IDs contain no hyphens; the models block uses the aliases prebuilt-analyzer-completion / prebuilt-analyzer-embedding; classify fields require an explicit enum array.

Step 2.3 — Test the samples via BINARY

- Submit each sample with begin_analyze_binary (no Blob, no SAS). Run the CU test harness; save the raw result JSON and the matrix scorecard.

- **B1 is settled, not tested:** the GA router response exposes no per-category classifier confidence, so gate B1 is dropped and field confidence (B4) is the only confidence gate.
- Verify `is_handwritten` returns yes on both handwritten samples and no on clean and scanned samples.

Step 2.4 — Accuracy scorecard

- One column per run (matrix format), with a run label. Confidence scores are LLM non-deterministic — values are stable but confidence varies run-to-run, so sample several runs before fixing any accuracy baseline.

Step 2.5 — Multi-page bug + GST gate

- Compare extracted vs actual line-item count on the multi-page sample. Confirm the GST gate (total minus GST reconciling to 5%, tolerance 0.02) routes inconsistent arithmetic to review.

Success criteria

- Classification correct on all samples (two-category router; non-invoices land in 'other').
- Clean digital invoices: critical fields correct, confidence ≥ 0.75 on 4 of 5.
- `is_handwritten` correct on all handwritten and all non-handwritten samples.
- B1 explicitly dropped; GST gate validated; scorecard completed.

Phase 3 — Processing ledger (Day 4 afternoon)

Status. Complete — validated by the ledger harness. This replaces v2's byte-hash store.

Step 3.1 — Table shape

- `InvoiceExtractProcessLog`: RowKey = sanitized SharePoint item UniqueId; PartitionKey = a short prefix of it. Insert and point-read a test row in Storage Explorer.

Step 3.2 — Idempotency and atomic claim

- Register an item id (Received); re-run the same id and confirm an idempotent update — no second row, original `IngestedUtc` preserved.
- Atomic claim: the first insert wins; a concurrent second insert for the same item id returns 409 and short-circuits. This is the only 'seen before' mechanism, and it is idempotency, not dedup.

Success criteria

- Seen item id (already decided) → skip and return the stored decision; unseen → claim then proceed.
- Point-read under 200 ms; idempotent re-runs create no duplicate row; atomic claim verified.

Phase 4 — Azure Function + Logic App skeleton (Days 5–7)

Status. The Function code is written and offline-validated (gates and all A1 branches pass). Remaining: deploy it, then build the Logic App that calls it.

Step 4.1 — Deploy the Azure Function (decision engine)

- Publish the function app; set app settings (AZURE_CU_*, AZURE_STORAGE_ACCOUNT, AZURE_TABLES_TABLE, AZURE_CLIENT_ID, FIELD_CONFIDENCE_THRESHOLD, A1_LEASE_SECONDS); assign the managed identity; confirm the two role assignments.
- Smoke-test with the bundled client against one sample PDF; confirm a decision is returned and a ledger row is written.

Step 4.2 — SharePoint trigger

- Logic App Standard; trigger 'When a file is created (properties only)'; polling 1–5 minutes — confirmed acceptable. Verify it fires on upload.

Step 4.3 — Fetch bytes and the item id

- 'Get file content' returns the bytes; read the item UniqueId. Base64-encode the content. No SHA-256, no Blob staging.

Step 4.4 — Call the Function

- HTTP POST to /api/process-invoice with sourceId (UniqueId), contentBase64, fileName, sourceFileUrl. Parse the returned decision. Gate A1, the CU call, and gates B2/B3/B4/GST all run inside the Function.

Step 4.5 — Branch on the returned decision

routingDecision	Logic App action
HAPPY_PATH_CANDIDATE	Write branch — Add a new row (Phase 6 wiring)
REVIEW_B3_HANDWRITTEN_OR_UNKNOWN / REVIEW_B4_CRITICAL_FIELD / REVIEW_GST_MATH	Review queue (Phase 5 wiring)
REJECT_B2_OTHER_CATEGORY / REVIEW_NO_CHILD_EXTRACTION	Review queue with the reject reason
alreadyProcessed = true	Do nothing (A1 short-circuit)

- The Logic App branches on the decision string only — it evaluates no confidence expressions. This is the main simplification over v2.

Success criteria

- The deployed Function returns a decision for a sample PDF; a ledger row is written (Received then Extracted).
- Upload triggers the Logic App, which calls the Function and branches on the decision.
- Clean digital invoice reaches the write branch; low-confidence and handwritten documents route to review.
- A re-fired trigger on the same item short-circuits at A1 (alreadyProcessed) with no second CU call.

Phase 5 — Human review queue (Day 8)

Step 5.1 — SharePoint review list

- InvoiceReviewQueue: InvoiceFileUrl, DocumentType, IsHandwritten, ExtractedFields (JSON), ConfidenceScores (JSON), RejectionReason, AnomalyFlag, Status (Pending/Approved/Rejected), AssignedTo, CreatedAt.

Step 5.2 — Wire queue writes

- Logic App 'Create item' in each review/reject branch; verify items appear with the firing gate in RejectionReason.

Step 5.3 — Minimal review interface

- Filtered list view (Pending). Reviewer edits the ExtractedFields JSON and sets Status = Approved. Power Apps deferred to implementation.

Step 5.4 — Approval Logic App

- Trigger 'When an item is modified' + Status == Approved. Parse the corrected JSON; wrap in a failure branch that resets Status to Pending on invalid JSON. Hand the normalised fields to the Phase 6 write action.
- The approval path re-enters the Logic App write action; it does **not** re-invoke the Function or CU, and there is no duplicate re-check.

Success criteria

- A review-routed document appears in the queue with its rejection reason.
- Approval triggers the second Logic App; malformed JSON resets to Pending instead of failing silently.

Phase 6 — Dynamics 365 CE write-back (Days 8–9)

Before you start. Prerequisites from pre-work: CE sandbox URL, invoice table logical name, field logical names, vendor/account lookup answer, service account or managed identity. Use the Dataverse connector — not the legacy Dynamics 365 connector. There is no duplicate query: per requirement, the happy path writes directly.

Step 6.1 — Connectivity test with hardcoded values

- Dataverse 'Add a new row' → sandbox → invoice table. Hardcode test values; run; verify the row appears in CE.
- If the invoice table requires an account lookup: create a test account first and pass its row ID — this validates the lookup-model answer.

Step 6.2 — Canonical field mapping

- Replace hardcoded values with expressions. Mapping (placeholder logical names): invoice_number → invoicenum; total_invoice_amount → totalamount; payment_due_date → duedate; vendor_name → name or account lookup.

Step 6.3 — Terminal ledger write after a confirmed row

- The ledger update to Status = Written runs only after 'Add a new row' succeeds ('Run after' = Succeeded only). Capture the returned row ID into DynamicsRecordId.
- A failed Dataverse write leaves the row without a terminal Written and without a DynamicsRecordId — the stuck-invoice signal.

Step 6.4 — Approval path

- The approved-review path converges on this same write action and writes directly (no dup re-check).

Success criteria

- Clean digital invoice flows end-to-end into a CE invoice row; ledger Written + row ID set after the 201.
- A failed Dataverse write does NOT write the terminal Written status (the row remains in-flight).
- The approval path writes successfully. No duplicate query anywhere (removed by requirement).

Phase 7 — End-to-end accuracy run (Days 9–10)

Step 7.1 — Full run, all samples

- Per document record: classified category vs expected, critical fields vs baseline, confidence scores, is_handwritten result, routed-to, routing correct?, CE row created, ledger Written.

Step 7.2 — Idempotency validation

- Same SharePoint item re-triggered (polling replay or manual re-run) → A1 short-circuits; no second CU call and no second Dynamics row.
- Two near-simultaneous triggers on the same item → the atomic claim lets exactly one proceed.

Out of scope (expected, not a defect). The same invoice re-uploaded as a NEW SharePoint file (new item id) or entered manually into Dynamics now creates a new Dynamics row. Duplicate-invoice verification is removed by requirement; record this behaviour as expected. The v2 re-exported-PDF and manual-entry-duplicate scenarios are dropped.

Step 7.3 — Multi-page line items + GST gate

- Compare the CE row line data vs the actual invoice; confirm or clear the arithmetic-check requirement. The GST gate routes inconsistent arithmetic to review.

Step 7.4 — Threshold calibration

- Per critical field: extraction rate, accuracy rate, confidence correlation. Adjust 0.75 if the data warrants. Report is_handwritten precision/recall. Account for confidence non-determinism by sampling multiple runs.

Step 7.5 — Findings document (one page, non-technical)

- Straight-through rate per document type; projected daily review-queue volume; multi-page bug impact; is_handwritten reliability; threshold recommendation; CE data-model decisions confirmed (account lookup); an explicit note that duplicate invoices are not verified (accepted) and the residual; go/no-go recommendation.

Success criteria

- Clean digital invoices: $\geq 80\%$ straight-through.
- Item-id idempotency behaves correctly (same upload $\rightarrow$ no duplicate row).
- Classification correct on all samples; at least one invoice completes the full path without intervention.
- Findings document delivered with a go/no-go recommendation.

Appendix — deferred to implementation

Deferred item	When to add
Property tax analyzer + a third router category	Post-prototype — the first planned extension
Duplicate-invoice detection (if ever re-required)	Explicitly de-scoped now; revisit only if the business reverses the requirement
Managed identity for Dataverse and SharePoint (replace service account / API key)	Before production — the prototype may use a key/service account
Power Apps review UI; Teams adaptive-card formatting	Implementation phase 1
Find-or-create account flow (if CE requires a lookup)	Implementation phase 1 — decided by the Phase 6 answer
Arithmetic validation layer	Implementation phase 1, if Phase 7 confirms the table bug
Error handling, dead-letters, retries	Implementation phase 1
Nightly ledger cleanup (expired rows; never stuck rows)	Implementation phase 2
ARM / Bicep templates, CI/CD	Before production
VNet / private endpoints; Dataverse throttling (429) monitoring	Before production / implementation phase 1